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 Studierichting: Informatica
 Oefeningenreeks 5J nr. 2.7

$$L = \int_0^4 \sqrt{(x'(t))^2 + (y'(t))^2} dt$$

$$x'(t) = \left(\frac{2}{3} t^{\frac{3}{2}} \right)' = t^{\frac{1}{2}}$$

$$(x'(t))^2 = t$$

$$y'(t) = \left(\frac{4}{9} t^{\frac{9}{4}} \right)' = t^{\frac{5}{4}}$$

$$(y'(t))^2 = t^{\frac{5}{2}}$$

$$L = \int_0^4 \sqrt{t + t^{\frac{5}{2}}} dt$$

$$L = \int_0^4 \sqrt{t(1 + t^{\frac{3}{2}})} dt$$

$$L = \int_0^4 t^{\frac{1}{2}} \sqrt{1 + t^{\frac{3}{2}}} dt$$

$$u = 1 + t^{\frac{3}{2}} \Rightarrow du = \frac{3}{2} t^{\frac{1}{2}} dt \Rightarrow t^{\frac{1}{2}} dt = \frac{2}{3} du$$

$$t = 0 \Rightarrow u = 1$$

$$t = 4 \Rightarrow u = 1 + 4^{\frac{3}{2}} = 1 + 8 = 9$$

$$L = \int_1^9 \sqrt{u} \cdot \frac{2}{3} du$$

$$L = \frac{2}{3} \int_1^9 u^{\frac{1}{2}} du$$

$$L = \frac{2}{3} \left[\frac{2}{3} u^{\frac{3}{2}} \right]_1^9$$

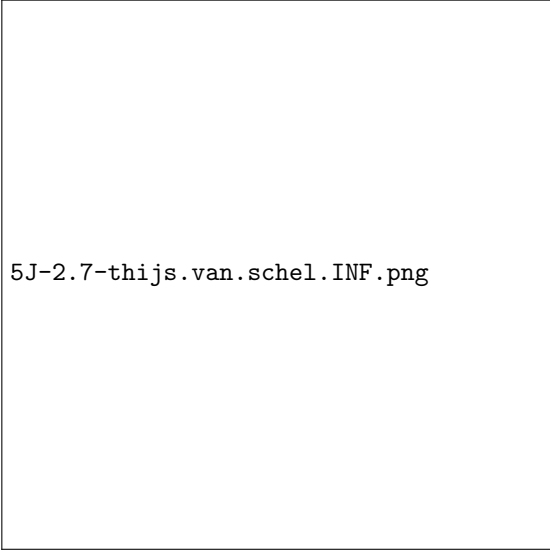
$$L = \frac{4}{9} \left(9^{\frac{3}{2}} - 1^{\frac{3}{2}} \right)$$

$$L = \frac{4}{9} (27 - 1)$$

$$L = \frac{4}{9} \cdot 26$$

$$L = \frac{104}{9}$$

tekening:



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References